

Department of Computer Science and Information Technology
Database Systems and Applications: CSIWZC337(WIMS 2024), SSWT
ZC337(WASE 2024), SEWI ZC337(SIM 2024)
Lab Assignment

1. Important Instructions:

- **Group Assignment:** This is a collaborative group assignment designed for teams of 5 members. Only one document needs to be uploaded per group. All students within a group will receive the same marks. You must explicitly mention the names and BITS-IDs of all group members on the first page of the submitted document.
- **Group Allocation:** The groups are pre-defined in Taxila portal and cannot be changed.
- **Format:** This is a theoretical, pen-and-paper assignment. No SQL coding or software tools are required.
- **Hand-drawn Requirement:** Students must draw the ER diagram and tabular schemas clearly by hand on a clean sheet of paper.
- **Digitization:** Scan or take a high-quality, legible photograph of your hand-drawn work and convert it into a single PDF file.
- **Submission Portal:** The completed assignment must be uploaded to the Taxila (eLearn) portal only.
- **Submission Status:** Assignments left in a "draft" state shall not be evaluated. Ensure you complete the final submission step.
- **Grading Weightage:** The evaluation will be marked out of an initial 20 marks. This score will then be converted/scaled down to a final weightage of **7 marks**.
- **Academic Integrity:** Refrain from copying or sharing documents with other groups. Any evidence of plagiarism or malpractice will attract 0 marks for all participating students.
- **Queries:** Any questions related to this assignment should be asked exclusively through the eLearn portal under the discussion forum.
- **Deadline:** The last date of submission is **2nd September 2026**.

2. Objective:

At the end of this assignment, students will be able to develop the skills necessary to translate business rules into a conceptual data model by:

- Identifying entities, attributes, and primary keys from a given problem statement.
- Determining the correct relationships and cardinalities (one-to-one, one-to-many, many-to-many) between entities.
- Designing a complete Entity-Relationship (ER) diagram.
- Mapping the conceptual ER diagram into a logical Relational Schema (drawing tables with primary and foreign keys).

3. Pre-requisite:

- Basic knowledge of conceptual database design and ER modelling concepts.
- Familiarity with standard ER diagram notations.
- Understanding of relational data models, primary keys, and foreign keys.

4. Lab Exercise:

Problem Statement: Integrated Railway Booking System

A railway organization operates several trains connecting different cities and wants to develop a Train Booking System to maintain information about trains, passengers, railway stations, train schedules, coaches, and ticket bookings.

Each train is identified by a unique train number and has a name and train type such as Express, Superfast, or Passenger. A train operates according to a schedule and may stop at multiple railway stations. A railway station can be served by multiple trains. For each train-station combination, the system maintains the arrival time, departure time, and stopping sequence number.

Passengers can book seats on multiple train journeys, and each train journey can have bookings from many passengers. Every booking record the booking date, seat number, coach number, fare, and booking status such as Confirmed, RAC, or Waiting.

Each train consists of one or more coaches. A coach belongs to exactly one train and is identified within that train by its coach number. Each coach contains multiple seats, and each seat is identified by its seat number within the coach.

A passenger can make multiple bookings, while each booking belongs to exactly one passenger and one train journey. The system should prevent the same seat from being assigned to two confirmed passengers for the same train journey.

The railway organization maintains details of stations, including station name, city, and state. A train can visit many stations, and a station can be visited by many trains over time.

The system should enable the railway organization to maintain details of passengers, trains, coaches, seats, railway stations, train schedules, and bookings. It should also support reports such as train-wise bookings, passenger travel history, station-wise trains, coach/seat availability, revenue generated from bookings, and train-wise passenger occupancy.

Business Rules

- One train can operate on multiple journeys, but each journey is associated with exactly one train.
- One train can have many coaches, but each coach belongs to exactly one train.
- One coach contains many seats, but each seat belongs to exactly one coach.
- A train can visit many stations, and a station can be visited by many trains.
- For each train-station combination, the system records the sequence number, arrival time, and departure time.
- A passenger can make many bookings, but each booking belongs to exactly one passenger.
- A train journey can have many bookings, and each booking is for exactly one train journey.
- Each booking is assigned to one seat and one coach when the booking is confirmed.
- The same seat cannot be assigned to two confirmed passengers for the same train journey.
- A booking must record booking date, fare, and booking status.

- A coach cannot exist without a train, and a seat cannot exist without a coach.
- A train journey must have a source station and a destination station, and the destination station must occur after the source station in the train's route.

5. Deliverables

Students are required to submit a single report (PDF format) containing the following:

a. Step-by-Step ER Modelling

- i. Develop the ER model systematically, showing intermediate steps instead of only the final diagram.
- ii. Include the identification of:
 1. Entities
 2. Attributes (simple, composite, multivalued, and derived, wherever applicable)
 3. Primary keys
 4. Relationships
 5. Cardinalities and participation constraints
 6. Min-max constraints (where applicable)
 7. Generalization/Specialization (if applicable)

b. Final ER Diagram

- i. Draw a neat and complete ER diagram using standard ER notation.
- ii. Ensure that all entities, attributes, keys, relationships, and constraints are clearly labelled.

c. ER-to-Relational Mapping

- i. Convert the final ER diagram into a relational schema by applying the standard ER-to-relational mapping rules.
- ii. Clearly specify:
 1. Relations (tables)
 2. Primary keys
 3. Foreign keys
 4. Mapping of strong and weak entities
 5. Mapping of different relationship types (1:1, 1:N, M:N)
 6. Mapping of generalization/specialization (if applicable)

d. Final Relational Schema

- i. Present the complete set of relations in a clear and readable format.
- ii. Clearly indicate the **primary key (PK)** and **foreign key (FK)** for each relation.

6. Marking Scheme (Total: 20 Marks)

(Note: Total score scales down to a final weightage of 7 marks.)

- **Step-by-Step ER Diagram (1 mark):** Systematically developed with intermediate steps and clearly stated assumptions.
- **Entities (2 marks):** Correctly identified, strong/weak distinguished, and no unnecessary entities introduced.
- **Attributes (2 marks):** Correctly assigned to entities with appropriate types (simple, composite, multivalued, derived) identified.
- **Primary Keys (1 mark):** Appropriate primary keys assigned, including correct partial keys for weak entities.
- **Relationships & Rules (4 marks):** Correctly identified cardinalities, participation constraints, min-max notation, and ISA hierarchy constraints.
- **Mapping of Entities (5 marks):** Strong and weak entities correctly mapped into relations with appropriate PKs and standard attribute mapping rules applied.
- **Mapping of Relationships (5 marks):** Correct mapping based on cardinality (1:1, 1:N, M:N), proper FK/bridge table placement, and reflection of participation/ISA constraints.